

Evan Joseph Gérard AYOUB

Robotics MSc student at EPFL specializing in computer vision, embedded systems and autonomous systems, with hands-on experience in vision-based mobile robots, aerial robotics and power electronics.

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Education

September 2025 – Present

MSc in Robotics, Minor in Data Science — EPFL

August 2023 – July 2024

Exchange Year — Aalto University, Helsinki

September 2022 – July 2025

BSc in Microengineering — EPFL | GPA: 5.00/6.00

September 2019 – July 2022

French Baccalaureate — Lycée Fénélon Notre-Dame, La Rochelle

Mathematics, Physics, Chemistry and Computer Science; European Section | 17.03/20, with honors



Selected Projects

Semester project | September 2026 – Present

Deep Learning for Visual Material Recognition on Edge Devices | DISAL, EPFL

Developing and evaluating deep-learning methods to classify materials from visual data, including benchmarking baseline models, investigating convolutional and transformer-based architectures, and deploying inference on resource-constrained edge devices.

MAKE project | September 2026 – Present

Space Robotics for the European Rover Challenge — Handling Device System Team | EPFL Xplore

Member of the Handling Device System (HDS) team within EPFL Xplore, developing Martian rover and drone systems for the European Rover Challenge.

October 2025 – June 2026

1st Place — Camera-Only Autonomous Robot at Robopoly Competition | EPFL

Built an autonomous robot from scratch using only visual input from an onboard camera for line tracking, corridor tracking and autonomous target shooting.

Innovative MAKE project | March 2024 – Present

MPPT & Power Electronics PCB for a Solar Aircraft | Sola Project, EPFL

Designed and assembled the power circuit PCB for a solar aircraft, with a focus on the maximum power point tracker.

Academic project | Grade: 6.00/6.00

Autonomous Thymio Navigation with Vision & Sensor Fusion | EPFL

Developed a Thymio navigation system using A* path planning, ArUco-based computer vision localization, EKF sensor fusion, ToF obstacle avoidance, kidnapped-robot recovery and automatic replanning.

Academic project | Grade: 5.25/6.00

High-Speed Autonomous Drone Gate Navigation | EPFL

Used Webots to develop gate-detection and navigation methods for flying through gates as quickly as possible, then worked in a team to transfer the approach to a real drone flying through physical gates.

Academic project | Grade: 5.75/6.00

Deep Learning for UNO Card Recognition & Player Attribution | EPFL

Detected UNO cards on a table, classified each card and attributed the detected cards to the correct player.

Academic project | Grade: 6.00/6.00

Statistical Data Analysis of Political Trading — Copycat Portfolios | EPFL

Analyzed financial data, selected an investment strategy and evaluated political trading performance, with the results delivered through a demonstration website.

Other Awards & Competitions

1st Place — Lions Club Physics Competition | 2022

Concours Général de Mathématiques, French National Competition — School-Selected Candidate | 2022

Experience

September 2023 – Present

Teaching Assistant and Mathematics/Physics Tutor | EPFL & Apprenons Ensemble

Tutor students in mathematics, physics and electronics across secondary, gymnasium and vocational levels.
Serve as a teaching assistant for Electrical and Electronic Systems and Mechanical Physics courses.

April 2022

Robotics Internship in R&D for mobile robotics | Shart Robotics

R&D workflow of a robotics company, from design and prototyping to testing.

September 2020 – July 2022

Programming and Mathematics Instructor | Lycée Fénélon Notre-Dame, La Rochelle

Introduced high-school students to Python programming and computational applications of mathematics.

Skills

Robotics & Autonomous Systems: Mobile robotics, legged robotics, autonomous navigation, path planning (A*), localization, sensor fusion (EKF), obstacle avoidance, trajectory tracking

Computer Vision, AI & Data: Computer vision, image processing, object detection and classification, deep learning, convolutional and transformer architectures, edge AI, statistical data analysis

Programming & Libraries: Python, C/C++, MATLAB, OpenCV, NumPy, Pandas, PyTorch

Engineering Tools: Git, Webots, MuJoCo, KiCad, LTspice, CATIA

Embedded Systems & Electronics: Embedded systems, PCB design and assembly, power electronics, MPPT, Raspberry Pi, Arduino, sensors, batteries and motors

Languages

French — Native | Lebanese Arabic — Native | English — C2 | Spanish — B2 | German — Beginner

Activities

Chief Scout | September 2021 – Present

Member, Association of the Lyrical Evenings of Sanxay | 2017 – Present

Interests: Handball, sailing, FPV drones, trumpet, guitar and judo (brown belt)